

Zhengyang Yu

Curriculum Vitae

San Diego & Bay Area

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Education

2025 – Jun 2028 **Bachelor of Science in Computer Science**, *University of California, San Diego*, San Diego, CA

Expected Graduation: June 2028

2023 – 2025 **Associate of Science in Computer Science**, *Laney College*, Oakland, CA

Projects

2026 **Quotaquarium**, *Personal Project*

- Designed and built a C++/FreeRTOS application that authenticates multiple accounts, retrieves live data over HTTPS, and turns changing quota data into an interactive physical display.
- Implemented concurrent network workers, adaptive polling, structured diagnostics, and modular UI state; investigated build, state-management, rendering, and hardware-integration failures through iterative testing.
- Protected sensitive data with validated TLS, device-code authorization, and bounded logs designed to exclude credentials, tokens, and raw API responses.

2025 **HTML to ePub Converter**, *Personal Project*

- Engineered a Python document-processing pipeline that parses HTML, normalizes content into a structured ePub package, embeds metadata, and preserves traceable source organization.
- Improved reliability through modular transformations, structured error handling, and repeatable validation of generated documents.

2024 **Home Assistant Presence Plugin**, *Personal Project*

- Built a Home Assistant plugin detecting real-time room occupancy using entry/exit sensor data.
- Designed event-driven architecture integrating Python scripts and RESTful APIs for smart lighting automation.

Skills

Programming C++, Python, Java, C

AI-Assisted Codex for codebase exploration, change planning, iterative implementation, debugging, and Development build/test verification; critical review of generated changes

Engineering Git, software testing, debugging, problem analysis, structured logging, technical documentation

Systems & Data Linux, Docker, HTTP/HTTPS, REST APIs, data processing, concurrent and event-driven systems, FreeRTOS

Tools PlatformIO, VS Code

Activities

2024–2025 Computer Consultant, Oakland Public Library - Elmhurst Branch

Supported community members with computer troubleshooting and taught basic software skills.

2024 Tutor, CTN (Community Technology Network)

Provided one-on-one tutoring to help adults improve digital literacy skills, including computer operation and internet usage.

2025–Present Member, UCSD Envision Makerspace